MIL-S-22308D
27 February 1987
SUPERSEDING
MIL-S-22308C
10 October 1980

MILITARY SPECIFICATION

SPRAYERS, INSECTICIDE, ELECTRIC-MOTOR-DRIVEN, HAND-CARRIED, ONE QUART TO ONE-AND-A-HALF GALLON CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-carried, electric-motor-driven, 1 quart (qt) to 1 1/2 gallon or 1 to 6 liter (L) capacity, mist-type and Ultra Low Volume (ULV) insecticide sprayers.

1.2 Classification. The sprayers shall be of the following types and rated capacities (see 6.2):

Type I - Standard.

Size 1 - 1 qt or 1 L.
Size 2 - 2 qt or 2 L.
Size 3 - 1 gallon or 4 L.
Size 4 - 1 1/2 gallons or 6 L.

Type II - ULV.

Size 1 - 1 qt or 1 L.
Size 3 - 1 gallon or 4 L.
Size 4 - 1 1/2 gallons or 6 L.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3740

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SPECIFICATIONS

FEDERAL

W-C-596 - Connector, Plug, Receptacle and Cable Outlet, Electrical Power, General Specification for.

MILITARY

MIL-S-12651 - Sprayers, Packaging of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting acitvity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Reg. No. 241-110-AA - Malathion ULV Concentrate Insecticide.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3092 - Particle Size Distribution of Space-Insecticide Aerosol.
E380 - SI (International System Units of Measurement).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 20 - General-Use Snap Switches.
UL 62 - Flexible Cord and Fixture Wire.
UL 73 - Motor-Operated Appliances.
UL 1004 - Electric Motors.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The sprayer shall consist of a housing, power unit, carrying handle, switch, electrical cord, dispensing system, and insecticide reservoir.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

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3.3 Standard commercial product. The sprayer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the sprayer being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5.1 Measurement systems. Unless otherwise specified (see 6.2), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, when measurements, dimensions, sizes and capacities are given in US units, these measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM E380.

3.6 UL listing. Each unit shall bear the Listing Mark of Underwriters Laboratories Inc.

3.7 Design. Design of the equipment shall be in accordance with the best engineering practice. The equipment design and accessory installation shall provide accessibility for maintenance and service in the field. Particular attention shall be given to features contributing to long periods of service without interruption for adjustment and repair. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment. The equipment shall conform to the insecticide sprayer requirements of UL 73.

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3.8 Housing. The housing shall enclose the spray mechanism and the power unit, and shall be compactly designed to provide for simultaneous carrying and spraying, and for spraying after placement of the sprayer on the floor. The housing shall be constructed of corrosion-resistant steel, aluminum alloy, or other rugged material, and shall be designed to withstand testing in accordance with 4.5.1 without damage and impairment of function. When tested in accordance with 4.5.8, metallic materials shall develop no evidence of electrolyte erosion, decomposition, metal transfer, or rust, and nonmetallic material shall develop no evidence of shrinking, swelling, warping, dissolving, softening, or any chemical change.

3.9 Carrying handle. The carrying handle shall be firmly attached in a convenient position for safe operation.

3.10 Switch. The switch shall be a snap action switch conforming to UL 20, with a rating not less than 15 amperes at 120 volts (V) alternating current (ac). The "ON" and "OFF" positions shall be clearly marked. The switch shall be conveniently located for safe operation while carrying the sprayer.

3.11 Cord. The cord shall conform to UL 62, type SJO or SJT or other type equally or more serviceable for the application, three-conductor, with not less than wire size No. 18 American Wire Gage copper conductors, and a 2-pole, 3-wire general-purpose plug conforming to W-C-596. Cord length shall be not less than 6 feet (1.8 meters (m)).

3.12 Power unit and dispensing system. The power unit shall be an electric motor conforming to NEMA MG 1 and UL 1004. Fungus-preventative treatment and radio interference supression may be furnished, but are not required unless specified (see 6.2) Unless otherwise specified (see 6.2), the motor shall have a nameplate voltage rating of 115V ac. The power shall be adequate for the performance (see 3.13). The dispensing system shall consist of a reservoir and spray mechanisms, with necessary connections to carry the insecticide from the reservoir to the spray mechanism. Removable strainers and screens shall be provided in the supply lines as required to prevent clogging of valves and spray mechanism. All parts shall be constructed, assembled, and supported, to withstand the type of intended usage without damage and impairment of function. Each internal part coming in contact with insecticide shall not change dimension, texture, or physical property, or show signs of chemical change.

3.12.1 Reservoir. The insecticide reservoir shall be constructed of AISI Grade 302 steel, insecticide-resistant aluminum alloy, or other insecticide-resistant material such as polyethylene. The reservoir shall be integrally constructed with the sprayer, or a nonreusable insecticide cartridge, and shall be provided with means for airtight connection in a manner to withstand dropping, bumping, and jarring without damage, detachment, jamming of the fasteners, or development of leaks. When filled to the operating level, the capacity of the reservoir shall be not less than the rated capacity, plus an allowance for air space.

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3.13 Performance.

3.13.1 Spray range. The unit shall make effective deposition of mineral oil at a spray range not less than 20 feet (6.1 m), when tested in accordance with 4.5.4.

3.13.2 Discharge rates. The unit shall be capable of a consistent flow rate which does not vary by more than 10 percent when tested in accordance with 4.5.5. The rate of flow may be fixed or variable, and shall be capable of at least one setting that is between 0.5 and 3 ounces (14.8 and 88.7 milliliters) per minute. A metal plate, or mylar aluminum type pressure-sensitive label, shall be accurately aligned to indicate the discharge rate with mineral oil.

3.13.3 Particle size. The sprayer shall discharge droplets with a Volumetric Mean Diameter (VMD) of 35 micrometers (µm) or less for type I, 20 µm or less for type II, and for both types an average large droplet size of 100 µm or less, when tested in accordance with 4.5.6.

3.13.4 Percent discharge. The sprayer shall discharge not less than 80 percent of the rated reservoir capacity when tested in accordance with 4.5.7.

3.14 Weight. The net weight of the sprayer shall not exceed 20 pounds (9.1 kilograms), when tested as specified in 4.5.2.

3.15 Noise limits. The noise level of the sprayer shall not exceed 95 decibles (A-weighted) (dB(A)) when tested in accordance with 4.5.9.

3.15.1 Precautions (noise). If the sound level exceeds 85 but does not exceed 95 dB(A), signs shall be provided which state, "CAUTION - HEARING PROTECTION REQUIRED". Hearing protectors shall not be supplied with the sprayer.

3.16 Reliability. The sprayer shall be capable of operating for 400 hours with a minimum Mean Time Between Failures of 100 hours.

3.17 Maintainability. The sprayer shall have a maintenance ratio of not more than 0.03 (see 4.5.11). Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair or replaced components and scheduled before-and-after operation checks are excluded.

3.18 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice.

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3.19 Treatment and finish. Unless otherwise specified (see 6.2), the equipment shall be treated and finished in accordance with the manufacturer's standard practice.

3.20 Technical publications. The contractor shall furnish to the Government repair parts list and operation and maintenance instructions normally furnished in commercial practice for each sprayer specified in the contract.

3.21 Item marking.

3.21.1 Identification marking. The equipment shall be marked for identification in accordance with the manufacturer's standard practice.

3.21.2 Caution marking. In addition to caution marking required by UL (see 3.6) and when hearing protection is required (see 3.15.1), each unit shall be marked as follows: "CAUTION - USE ONLY UNDER THE DIRECT SUPERVISION OF A CERTIFIED PEST CONTROL OPERATOR. USE PERSONAL PROTECTION REQUIRED BY OSHA, 29 CFR 1910.132, 1910.133, AND 1910.134, AS APPLICABLE. OCCUPIED AREAS MUST BE EVACUATED AND CLEARLY POSTED, PRIOR TO OPERATION." Marking shall be solvent- and wear-resistant so that it remains legible during the expected use-life of the unit.

3.22 Replacement parts. When specified (see 6.2), the following list of high-mortality items shall be included for minor field repair:

a. nozzles
b. gaskets
c. bands and clamps
d. o-rings
e. hoses
f. electric motor or brushes

3.23 Workmanship. Workmanship shall be in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one sprayer when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the test conditions and rejection criteria of 4.5, the tests of 4.5.1 through 4.5.4, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete sprayer. All sprayers of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

4.4 Examination. Each sprayer selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.5 Testing. The maintenance schedule of the operator's manual shall be followed during the testing sequence of 4.5.1 through 4.5.10 to determine compliance with 4.5.11. Failure to pass any phase of any test shall be cause for rejection. Leaks, loss of insecticide or chemical solutions, evidence of electrolytic action and chemical changes, and any uncorrectible defect shall be causes for rejection of the inspection lot.

4.5.1 Ruggedness. Drop the empty sprayer three times, from a height of 3 feet (0.9 m), onto a hard wood surface. The manner of dropping shall be such as to cause the sprayer to strike on entirely different positions in the three drops. Examine the sprayer to determine damage, such as cracks, which would affect operation of controls, cause leaks, loosen handle and reservoir connections, break or misalign tubing, sprayer mechanism, and motor bearing supports, expose electrically charged parts to accidental contact, and any other damage or distortion which could affect function in accordance with 3.13.

4.5.2 Weight. Weigh the empty sprayer to determine conformance to 3.14.

4.5.3 Capacity. Add the rated capacity volume of mineral oil to the empty sprayer reservoir. Overflow, or rise in liquid above the operating level (see 3.12.1), shall be causes for rejection.

4.5.4 Spray range. Charge the sprayer with mineral oil and position in a large room. Place four microscope slides, coated with polytetrafluoroethylene 1/, or silicone coating (see 4.5.6), 20 feet (6.1 m) from the spray mechanism or jet discharge nozzle. Operate the sprayer for not longer than 5 minutes with the discharge directed at the slides, and examine the slides for evidence of particle deposits. Absence of unmistakable visible evidence of spray impact on all of the slides shall be cause for rejection. If the sprayer cannot be repaired and pass a retest, the lot shall be rejected.

1/ Teflon, Registered Trademark of E.I. duPont de Nemours & Co., Inc., or equal. The source of prepared slides will be available from the contracting officer.

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4.5.5 Discharge rates. Following the test of 4.5.4, add a measured volume of mineral oil to the reservoir and operate for a fixed period of time. After completion of the dispersal period, measure the remaining volume of liquid and determine the discharge flow rate. Perform the test three times. Record individual values to determine conformance to 3.13.2. If the machine is equipped with a variable flow rate control, the test shall be conducted at the low and high discharge rate settings.

4.5.6 Particle size. Following the test of 4.5.5, charge the sprayer with mineral oil at 75 ± 10 degrees (°) Fahrenheit (23.9 ± 5.6° Celsius) and start spraying. At a distance of 4 feet (1.2 m) from the spray mechanism or nozzle, and at right angles to the jet discharge, rapidly pass a coated microscope slide through the spray using the maximum flow rate. During the pass, the coated side of the slide shall be perpendicular to the ground, with the coated surface facing the aerosol cloud. The slide coating shall be polytetrafluoroethylene or, as an alternate where required, silicone resin prepared as specified in ASTM D3092. Pass the slide through the aerosol cloud as many times as necessary to obtain sufficient droplet density for convenient measurements. By means of a 400-power or greater magnification microscope equipped with a stage micrometer disk, measure the diameters of a sample of 200 droplets. Determine the droplet sizes using the procedure for particle selection given in the EPA label requirements for Malathion ULV 2/ Concentrate Insecticide, section III. Use a spread factor of 0.6 for Teflon-coated slides exposed to mineral oil. Repeat the procedure to give a total of three sample slides. For each test slide, calculate the VMD, and the average diameter of the largest three droplets in the 200 droplet sample. Record individual values to determine conformance to 3.13.3.

4.5.7 Percent discharge. Following the test of 4.5.6, fill the sprayer reservoir with mineral oil to the operating level and operate at the maximum discharge rate until all discharge spray has stopped. The remaining oil shall be measured to determine conformance to 3.13.4.

4.5.8 Insecticide resistance. Following the test of 4.5.7, all replaceable and serviceable parts of the sprayer shall be disassembled. The insecticide reservoir shall be filled with one accurately measured rated capacity volume of 4.65 percent dischlorvos (DDVP), tightly sealed, and stored for a period of not less than 72 hours. All other parts which come in contact with insecticide, during normal operations, shall be immersed, for a period of not less than 72 hours in each solution, in the following test solutions consecutively:

a. Three percent salt solution.
b. Salt solution with 1.5 percent malathion emulsion made from 57 percent concentrate.
c. Undiluted deobase (No. 9 oil).

2/ Trademark of American Cyanamid Company.

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The parts and the used test solutions shall be examined (see 3.12 and 3.12.1) to determine evidence of electrolytic action and chemical action which might impair safe operation. The DDVP in the reservoir shall be measured by level or weight to determine any loss. The sprayer shall then be reassembled and tested for spray range in accordance with 4.5.4. Any loss of DDVP during storage shall be cause for rejection of the inspection lot. Any damaged part prior to reassembly shall be cause for rejection and replacement of the part tested. Ineffective performance after reassembly shall be cause for rejection of the inspection lot.

4.5.9 Noise limits. Following the test of 4.5.8, measure the noise level at the ear of the operator when the machine is held in a normal operational posture. The sprayer shall be operated at the maximum noise condition. A test measurement which exceeds 95 dB(A) shall constitute failure to pass this test.

4.5.9.1 Hazard precautions. Should the noise level exceed 85 dB(A) at the ear of the operator, failure to provide the precautions specified in 3.15.1 shall constitute failure to pass this test.

4.5.10 Reliability. Following the noise level tests, begin the 400-hour reliability test. During the testing, the machine shall be shut off after every 2 hours of operating time. The unit shall not disperse mineral oil except at the 200-hour point and the 400-hour point. At these times, the sprayer will be charged with mineral oil and tested for droplet size in accordance with 4.5.6. The overall failure criteria for the reliability testing is as follows: If three or fewer failures occur, the unit shall be considered acceptable. A failure for the purpose of this test is defined as any malfunction which causes inability to commence operation, cessation of operation, or degradation below the operating conditions required to continue the test, or causes a serious safety hazard of personnel or equipment.

4.5.11 Maintainability. Maintenance operations shall be performed during the 400-hour reliability test to determine conformance to 3.17.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-S-12651. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-S-12651 with the level of preservation and the level of packing as specified (see 6.2).

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6. NOTES

6.1 Intended use. The hand-carried, electric-motor-driven sprayers are intended primarily for indoor control of flying pests by dispersing water-based emulsions and oil solutions of insecticides. Type I sprayers are for general mist-use. Type II ULV sprayers are for use when it is necessary to avoid wetting surfaces, such as food handling areas or fabric surfaces, or for penetration of crevices to flush out insects.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size of sprayer required (see 1.2).
c. When specific measurement system is required (see 3.5.1).
d. When a first article inspection is required (see 4.2.1 and 6.4).
e. When fungus-preventive treatment and radio interference suppression is required (see 3.12).
f. Voltage of electric power supply if other than as specified (see 3.12).
g. When treatment and painting shall be other than as specified (see 3.19).
h. When optional high-mortality item repair parts are to be supplied (see 3.22).
i. When a preproduction pack inspection is required and time frame required for submission (see 4.2.1 and 4.6).
j. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Insecticide
Sprayer

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Review Activities:
Army - MD
DLA - CS

Preparing Activity:
Navy - YD

(Project 3740-0120)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-22308D	SPRAYERS, INSECTICIDE, ELECTRIC-MOTOR-DRIVEN, HAND-CARRIED, ONE QUART TO ONE-AND-A-HALF GALLON CAPACITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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